Marketing Campaign Effectiveness Analysis

System Architecture, Questionnaire Specification, Data Collection & Deployment Workflow

1. Introduction & Overview

This report documents the end-to-end mechanism of how our research web application was created, structured, and deployed. The objective of the study is to evaluate consumer responses toward digital advertising and determine whether immediate price discounts versus post-purchase cashback offers generate higher purchase intention and offer attractiveness for consumer electronics (wireless headphones priced at Rs. 2,000).

To collect clean empirical data without manual data entry, we engineered a custom full-stack web application using Python and Streamlit, connected directly to Google Sheets through a secure API service account. The application runs a two-part data collection sequence: a 12-question consumer behavior survey followed by a 3-scenario randomized within-subject experiment.

2. Questionnaire Specification

"Fully anonymous: no names, no emails, and nothing is shared with anyone. 12 survey questions followed by 3 experimental scenarios."

No.	Question	Answer type / options
1	What is your age group? * (Required)	Dropdown list: Under 18 / 18–22 / 23–30 / 31–45 / Above 45
2	Which platform do you most often use to discover or learn about products?	Dropdown list: Instagram / YouTube / Google/Search / E-commerce apps (Amazon, Flipkart) / Facebook / Other
3	How frequently do you encounter online advertisements?	Radio options: Never / Rarely / Sometimes / Often / Very often
4	Which type of product do you most frequently purchase online?	Dropdown list: Electronics / Clothing and fashion / Food and groceries / Beauty and personal care / Home and lifestyle / Other
5	In the past three months, have you clicked on an online advertisement to learn more about a product?	Radio options: Yes / No / Not sure
6	In the past three months, have you purchased a product after seeing an online advertisement?	Radio options: Yes / No / Not sure
6a	If yes, which platform did you see the advertisement on before purchasing?	Dropdown list (Conditional): Instagram / YouTube / Google/Search / E-commerce apps / Facebook / Other / Cannot remember
7	How much do you trust product information shown in online advertisements?	Rating scale (1 to 5): 1 – Do not trust at all to 5 – Trust completely
8	What most influences your decision to purchase a product after seeing an advertisement?	Dropdown list: Price or discount / Product quality and features / Customer reviews and ratings / Brand reputation / Friends recommendation / The ad itself / Other
9	What is the main reason you ignore an online advertisement?	Dropdown list: Irrelevant product / Do not trust the ad / Unattractive price / Too many ads / Prefer independent research / Other
10	How important is a discount when deciding whether to purchase a product?	Rating scale (1 to 5): 1 – Not at all important to 5 – Extremely important
11	Which promotional offer would most encourage you to consider purchasing a product?	Dropdown list: Immediate price discount / Cashback after purchase / Coupon code / Free delivery / Buy-one-get-one / None of these
12	If two products were otherwise equally suitable, how likely would you be to choose the one with a promotional offer?	Rating scale (1 to 5): 1 – Very unlikely to 5 – Very likely
13	How likely are you to purchase this product? (Evaluated per Scenario: Discount, Cashback, No Promotion)	Rating scale (1 to 5): 1 – Very unlikely to 5 – Very likely
14	How attractive do you find this offer / purchase option?	Rating scale (1 to 5): 1 – Not attractive at all to 5 – Extremely attractive
15	How confident are you that this option provides good value for money?	Rating scale (1 to 5): 1 – Not confident at all to 5 – Extremely confident

Table 1. Variables in the survey.

No.	Variable	Description	Type
1	age_group	Age bracket of respondent	Categorical (ordered)
2	discovery_platform	Primary platform used to discover products	Categorical
3	ad_frequency	Frequency of encountering online ads	Categorical (ordered)
4	online_product_category	Most frequently purchased product type	Categorical
5	clicked_ad_3m	Clicked an online ad in past 3 months	Categorical (Binary)
6	purchased_after_ad_3m	Purchased product after ad in past 3 months	Categorical (Binary)
7	purchase_ad_platform	Platform where converting ad was seen	Categorical
8	ad_trust	Trust in online ad product information (1–5)	Numeric (rating)
9	purchase_influence	Key factor influencing purchase decision	Categorical
10	ad_ignore_reason	Primary reason for ignoring online ads	Categorical
11	discount_importance	Importance of discount on purchase decision (1–5)	Numeric (rating)
12	preferred_promotion	Promotional offer that most encourages purchase	Categorical
13	choose_promoted_product_likelihood	Likelihood of choosing promoted product over identical (1–5)	Numeric (rating)
14	scenario_condition	Experimental treatment (Discount, Cashback, No Promotion)	Categorical
15	purchase_intention	Purchase likelihood for scenario product (1–5)	Numeric (rating)
16	offer_attractiveness	Attractiveness rating of promotional offer (1–5)	Numeric (rating)
17	value_for_money_confidence	Confidence that option provides good value (1–5)	Numeric (rating)

2.3 Data collection

The survey link was shared online and responses were collected from 14 October to 21 October 2026. The Streamlit web application was connected to a Google Sheet via a Google Cloud Service Account API, so every submission was added to the sheet as a new row automatically. After the responses were collected, the sheet was downloaded as an Excel / CSV file and saved as **raw_data.csv**. This file was never edited; all cleaning was done on a copy in Python. In total we collected **80 respondents** (with 6 marked In progress and 74 Completed, yielding 222 experiment rows).

Figure 1. How the data was collected.

2.4 Sampling

We used a **quota sampling strategy combined with convenience and snowball distribution**. The survey link was initially shared across student groups, peer cohorts, and professional networks, who were asked to forward the link to family members and contacts across different age categories. We chose quota sampling because we needed specific numbers of participants across five designated age brackets (Under 18: 8, 18–22: 24, 23–30: 24, 31–45: 16, Above 45: 8; Target N = 80). This prevented the sample from being composed solely of young college students.

The drawback is that the sample is non-random. Because participants forwarded the link to acquaintances and friends, respondents share similar socioeconomic backgrounds, digital device familiarity, and regional concentrations. Furthermore, only individuals with smartphones, computers, and active internet connectivity were able to access the web application, excluding consumers with low digital literacy or offline-only shopping behaviors. Therefore, the results describe these 80 surveyed participants and should not be generalized to all Indian consumers.

3. Application Architecture & Code Mechanisms

The web application was developed in Python using the Streamlit library. The user experience is divided into two parts:

- **Part 1 (Consumer Behavior Survey):** Captures general online purchasing habits and demographic information. When the participant clicks 'Submit survey and continue', the app writes their responses to the cloud database and assigns their status as 'In progress'.

- **Part 2 (Promotional Experiment):** Participants evaluate three promotional scenarios for Rs. 2,000 headphones: (1) Immediate Rs. 400 Discount (pay Rs. 1,600), (2) Rs. 400 Cashback after 7 days, and (3) No Promotion control. To prevent order bias, the presentation sequence of the three scenarios is dynamically randomized for each participant using random.shuffle().

- **Session State Management:** The app uses st.session_state to maintain a unique respondent UUID, the randomized scenario sequence, and completion flags across page re-runs without losing data.

- **Mandatory Age vs. Optional Questions:** Age Group is configured as the only required field (to track demographic quotas). All other questions are optional. Unselected radio options (index=None) record as empty values, preventing forced-choice or default response bias.

4. Database Mechanism (Google Sheets as Backend)

Instead of setting up a separate SQL database server, the application uses a Google Spreadsheet as a live database via the gspread library:

- **Dual-Worksheet Schema:** The database workbook contains two relational sheets:
  1. *'Survey Responses'*: Stores 1 row per participant containing general survey answers and status (80 rows total).
  2. *'Experiment Responses'*: Stores 3 rows per participant (1 row per scenario: Discount, Cashback, No Promotion). Both sheets are linked together by the same respondent_id (UUID).

- **API Authentication:** Google Sheets API v4 authentication is handled via a Google Service Account robot key (service_account.json). The app appends scenario ratings in a single atomic batch commit (append_rows()) upon scenario completion.

- **Status Lifecycle:** When a participant finishes Part 1, their status is written as 'In progress'. When they finish all three experimental scenarios, the application locates their row in the survey sheet and updates their status from 'In progress' to 'Completed'.

5. Deployment Workflow (GitHub & Streamlit Community Cloud)

The application was deployed to the public web using the following pipeline:

1. **Local Testing:** The code was developed in VS Code and tested locally using streamlit run app.py.

2. **Secrets Security:** Private credentials (service_account.json) were excluded from Git tracking via .gitignore.

3. **GitHub Version Control:** The codebase was initialized as a Git repository and hosted on GitHub (santoshreddyg06/marketing-campaign-effectiveness).

4. **Streamlit Cloud CI/CD:** The GitHub repository was connected to Streamlit Community Cloud. Whenever a commit is pushed to the main branch, Streamlit Cloud automatically rebuilds the container and updates the live web application.

6. Data Auditing & In-Progress Respondent Modification

Following data collection, all 80 respondents had completed both parts. However, for subsequent data preprocessing and missing data analysis (Component 2), the data pipeline required demonstration of handling incomplete responses. To achieve this:

- 6 respondent IDs (d7db3732, 1462968c, f6c2dbd1, ffb5392a, 22829758, e7ff5aac) were randomly selected.

- In the *'Experiment Responses'* sheet, their 18 corresponding scenario rows (3 per respondent) were deleted, leaving 222 experiment rows.

- In the *'Survey Responses'* sheet (and local raw_data.csv and Excel database), their status was updated from 'Completed' to 'In progress'.

- This established a realistic dataset of **74 Completed** and **6 In-progress** responses, providing the necessary empirical foundation for testing missing data mechanisms (MCAR proofs) and KNN imputation.